Car control

7a

In this activity we are using graphs to analyse Chris's off-road driving test. First we are developing a distance–time graph, then a speed–time graph, and finally an acceleration–time graph.

As part of preparing for her licence, Chris had the opportunity of developing her car control through a series of off-road driving tests devised by her instructor. Her instructor placed a line of witches hats 100 metres apart along the edge of the driving path. Smaller markers were positioned midway between successive witches hats. Chris's instructions were as follows.

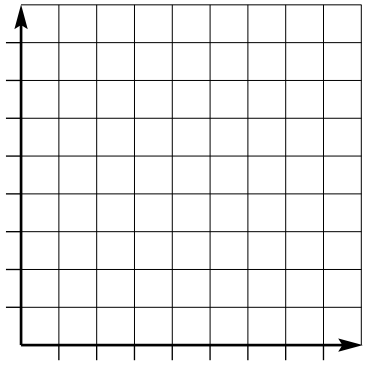
1. Start at zero time at the first hat and move with smooth acceleration to the first small marker in 10 seconds.
2. Drive on at constant speed past the second hat to the second small marker in the next 10 seconds.
3. Brake smoothly to stop at the 200 m mark (3rd witches hat) in 10 seconds.
4. Wait for 10 seconds.
5. Drive smoothly to the next hat in 15 seconds.
6. Brake over 10 seconds to come to rest in 50 metres.
7. Discuss your skill development in the stationary car for 10 seconds.

Developing a distance–time graph

1 **a** Enter the time and distance data from the instructions for the test in the table below.
b Label the axes of the graph grid provided and mark points on the graph to correspond with the data in your table. (Place time on the horizontal axis.)
c Notice that there are seven sections to the graph. Label these sections from A through to G.

Time (s)	Distance (m)

Distance–Time Graph



2 Look at your graph and answer these questions.

a What is the car doing in the graph segments where the graph is horizontal?

_____

b What is happening with the car during the graph segments where the graph is turned upwards?

_____

c When the curve is steepest, what is happening with the car?

_____

d What is happening with the car during the graph segments where the graph slopes a little?

_____

e The slope of the curve tells you at what speed the car is travelling. Speed is the change in distance in a certain time. In which section (A to G) is the speed:

– zero? _____ – lowest? _____ – highest? _____ – uniform but not zero? _____

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Developing a speed–time graph

Speed is the change in distance over a certain time.

3 **a** What is the formula for finding speed? _____

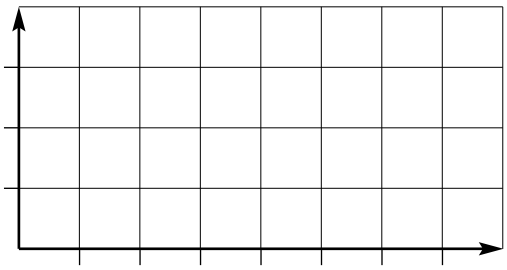
The table below provides space for entering data and calculating the speeds in the different sections of the test drive. Values have been shown in the first line of the table, and the average speed calculated for the starting segment of the test. Below the table there is an explanation of the entries made for this initial line.

Section	First time t_1 (s)	Second time t_2 (s)	Time $t_2 - t_1$ t (s)	First distance d_1 (m)	Second distance d_2 (m)	Distance $d_2 - d_1$ d (m)	Distance / time calculation	Average speed (m/s)	Final speed (m/s)
A	0	10	10	0	50	50	50/10	5	10
B	10								
C	20								
D	30								
E	40								
F	55								
G	65								

Section A: The time interval starts at time zero and extends for 10 seconds. The time length for Section A is then $t_2 - t_1$ = 10 – 0 = 10 s. During that time the car moved from 0 m to 50 m, i.e. it moved 50 m – 0 m = 50 m ($d_2 - d_1 = d$).
The average speed is obtained by dividing the distance travelled by the time taken ($v = d/t$), i.e. 50 m/10 s to give an average speed of 5 m/s. Note that at 10 s the final speed is 10 m/s. The 5 m/s speed is the average of the initial speed 0 m/s and the final speed 10 m/s for the section. Refer to Chris's instructions for the final speed.

b Complete the table above. Enter the known times and distances and calculate the speeds for the different sections of the test.

c Take the data on final speeds from the above table and mark the final speed, second time coordinates on the graph below. Make sure you label the axes with titles, units and numbers as appropriate.



4 Look at your speed–time graph and answer the following questions.

a When the curve in the graph is horizontal, what is the car doing?

_____

_____

b When the curve is sloping upwards, what is the car doing?

_____

c When the curve is sloping up most steeply, what is the car doing?

_____

d When the curve is sloping up but not most steeply, what is the car doing?

_____

e When the curve is sloping downwards, what is the car doing?

_____

f When the curve is sloping downwards most steeply, what is the car doing?

_____

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g The curve can be divided into seven sections. Each section is a straight line. Start from the left and label the sections A, B, C, D, E, F, G. Note that section A goes from 0 s to 10 s (at 10 s the car has reached a speed of 10 m/s). Section B covers the car's speed from 10 s to 20 s.

h The slopes of the curve show what accelerations the car experiences. Acceleration is the rate of change in speed in a certain time.
In which section (A to G) is the acceleration (i.e. positive acceleration):

– zero? _____ – lowest? _____ – highest? _____

In which section is the deceleration (i.e. negative acceleration):

– smallest? _____ – greatest? _____

Developing an acceleration–time graph

Acceleration is the change in speed over a certain time.

5 **a** What is the formula for finding accleration? _____

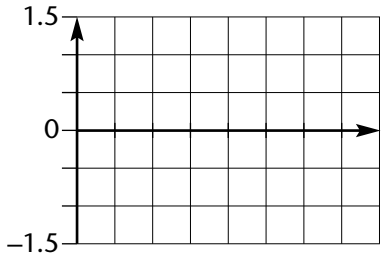
The table below provides space for entering data and calculating the accelerations in the different sections of the test drive. Below the table there is an explanation of the entries made for A, the first section of the test.

Section	Initial time t_1(s)	Final time t_2(s)	Time $t_2 - t_1$ t (s)	Initial speed v_1 (m/s)	Final speed v_2 (m/s)	Speed $v_2 - v_1 = v$ (m/s)	Speed/ Time calculation	Average acceleration (m/s^2)
A	0	10	10	0	10	10	10/10	1
B	10							
C	20							
D	30							
E	40							
F	55							
G	65							

Section A: The time interval starts at time zero and extends for 10 seconds. During that time the car's speed went from 0 m/s to 10 m/s. The acceleration during this period is given by the change in speed divided by the time for the speed change: 10 m/s/10 s = 1 m/s^2. For the first 10 seconds of the test the average acceleration was 1 m/s^2.

b Complete the table above about acceleration. Enter the known times and speeds and calculate the accelerations for the different sections of the test.

c Take the data on acceleration from the above table and mark the acceleration–time coordinates on the graph at right. Label the graph axes with a title, units and numbers.



6 **a** When sections of the curve are in the positive, what is the car doing?

_____

b When sections of the curve are on the x-axis, what is the car doing?

_____

c When sections of the curve are in the negative, what is the car doing?

_____

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